

Prototyping Kodi: Defining Design Requirements to Develop a Virtual Chat-bot for Autistic Children and Their Caregivers

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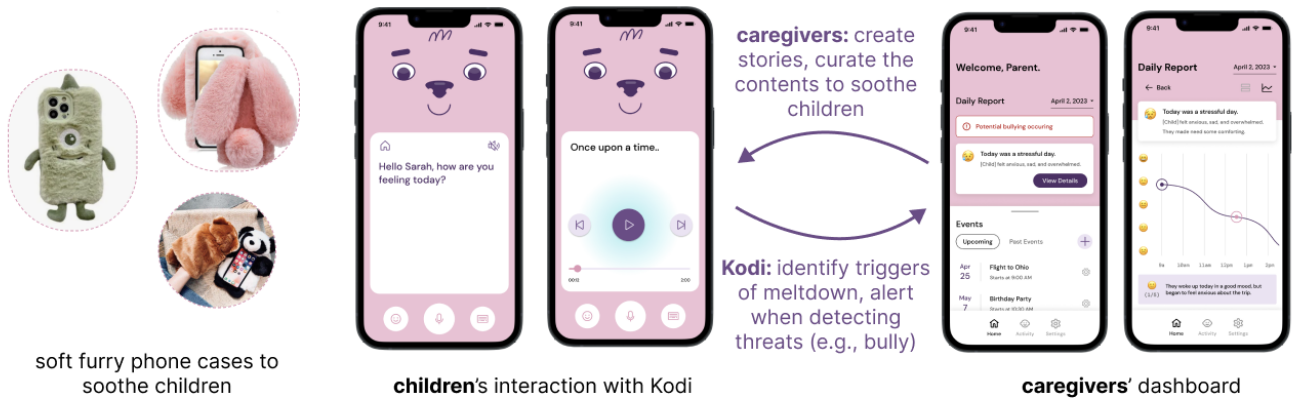


Figure 1: Kodi: a virtual conversational chatbot that connects autistic children and their caregivers

ABSTRACT

Autistic children's emotional meltdown is a common challenge faced by their caregivers, including their teachers and families. Communicating their problems can be extremely challenging for many autistic children, making it difficult for caregivers to identify the triggers of these meltdowns. Previous studies have revealed that autistic children find it easier to communicate with non-human objects, such as comfort dolls or social robots. In light of this, we have designed Kodi, a mobile virtual conversational agent aimed at bridging the communication gap between autistic children and their caregivers during distressing situations. Kodi's appearance can be personalized to match the child's favorite characters, such as a bunny or a dinosaur. To provide additional comfort, Kodi comes with furry phone cases that offer haptic stimulation. When the child faces stressful situations, caregivers can prompt Kodi on children's mobile device, allowing the child to express their emotions with

various input modalities (emojis, text, voice) and listen to soothing stories. Caregivers can tailor these stories to suit each child's unique needs. Furthermore, Kodi will employ keyword analysis to summarize conversations, identify potential triggers for emotional meltdowns (e.g., loud noises in a shopping mall, unexpected changes at school), and alert caregivers of potential problems, such as bullying. The data collected through Kodi holds the potential to be leveraged for predicting emotional meltdowns using machine learning techniques. This paper discusses the implications of the Kodi platform and outlines future research directions in this area.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools**; *Human computer interaction (HCI)*.

KEYWORDS

autism, social chat bot, conversational agent

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1 INTRODUCTION

Autism spectrum disorder (ASD) is a pervasive developmental disorder characterized by social and communicative impairments [6]. The Centers for Disease Control and Prevention (CDC) estimates that approximately 1 in 36 children are affected by ASD in the United States [13]. The daily lives of autistic children are often accompanied by a range of challenges, including communication difficulties, social challenges, and sensory overloads [18, 22]. The changes in daily routine and sensory stimuli can often trigger anxiety and distress for autistic children, which often results in their emotional meltdown [22]. The manifestation of children's emotional meltdowns may involve the refusal to communicate or interact, withdrawal, shutting down, or self-harming behaviors, exacerbating the difficulty of identifying the triggers of such behavior [15, 22]. The communication difficulties faced by children with ASD impede their ability to articulate their emotions and the reasons for their distress, which consequently delays appropriate responses from caregivers and educators. This has a significant impact on these children and their families.

This study aims to tackle this issue by designing a mobile application that facilitates communication between caregivers and children with ASD, with the ultimate goal of aiding caregivers in identifying the triggers of emotional meltdowns and enhancing their ability to manage them. Previous studies suggest that children with ASD tend to prefer interacting with inanimate objects like toys or comfort dolls rather than with humans [3, 14, 20, 29]. In light of this, we designed a virtual conversational agent named Kodi that helps children communicate their struggles. First, caregivers can schedule prompts for Kodi on their children's mobile devices when they expect them to encounter stressful situations, such as traveling, going to the hospital, or starting a new school. Second, Kodi will use emojis and pictures to ask children about their current emotional state, and children can respond either verbally or through text messages. Kodi will then provide a soothing story or suggest playing their favorite games based on the child's inputs. Caregivers can customize the content by writing their own stories or selecting games that are most appropriate for their child, thus facilitating indirect communication between the caregiver and child. Lastly, Kodi will analyze the text to identify potential triggers for a child's meltdown, such as loud noise or bullying at school, and alert caregivers if it detects any potential threats to the child.

Our work has made several contributions to the design of an application for the autism community. To gain insights into the diverse experiences of children and their caregivers and to identify ongoing issues, we utilized social media platforms such as Facebook parent groups, Reddit, and YouTube video logs. We thoroughly reviewed hundreds of posts to better understand the perspectives of different families. Through our qualitative data analysis, we found that one of the most pressing challenges faced by teachers and parents is handling emotional meltdowns and communication shutdowns in children. Previous studies have primarily focused on using virtual agents or social robots to improve social skills in autistic children, like training them to make more eye contact or providing speech therapy [23, 26]. However, our study takes a different approach. Our aim is to utilize virtual agents to facilitate

meaningful connections between children with autism and their caregivers, with a specific focus on managing emotional meltdowns.

This paper is structured as follows: First, we discuss the related literature on virtual social agents and examine existing works on communication tools for autistic children and their caregivers. Second, we present our preliminary studies, which explore recent posts made in online Facebook autism communities. Third, we present our design process for the Kodi application. Finally, we conclude this paper by discussing future research directions.

2 RELATED WORKS

The use of virtual social agents, such as conversational chatbot, avatars, and social assistive robots, has become an increasingly popular approach to supporting the social communication of autistic children [14, 24, 26]. These virtual agents can offer a predictable and non-judgmental environment for communication and interaction for autistic children, minimizing potential anxiety and stress that may be associated with human interaction [3, 20, 29]. Several studies have reported that autistic children tend to prefer virtual agents to humans. For instance, Diehl et al. [4] investigated the preferences of autistic children for interacting with a robot versus a human therapist and found that the children spent significantly more time interacting with and looking at the robot, and also exhibited more social and communication behaviors in the presence of the robot compared to the human therapist. A systematic review by Pennisi et al. [14] provided empirical evidence that supports positive interactions between virtual agents and autistic children across various contexts. The review revealed that ASD children performed better in collaboration with robot partners compared to human partners [30], had more social interaction with robots [27], and engaged more in social activities when the robot was used as a medium for storytelling as compared to human readers telling the same story [29]. While a body of literature suggests the benefits of virtual agents for children with autism spectrum disorder, the majority of studies have primarily focused on therapeutic interventions and enhancing social skills [1, 3, 14, 16, 19]. Spiel et al. [26] conducted a review of 185 papers that explored technology development for autistic children, and discovered that nearly half of the existing research focused on social skills training. Their systematic review reveals a lack of research aimed at improving the well-being of autistic children, highlighting the need for further studies that address the challenges faced by both caregivers and autistic children in promoting emotional well-being.

Autistic children's emotional meltdowns can be a challenging experience for both caregivers and teachers, as children often refuse to communicate under extreme stress, making it even more difficult to take appropriate actions. Studies found that use of virtual agents and social assistive robots are effective solutions in reducing anxiety and improving emotional regulation skills [2, 3]. For instance, Loi et al. [12] investigated the use of a social robot named Betty and found that Betty reduces stress and comfort neuropsychiatric patients through conversations, relaxation exercises, and games. Similarly, Vanderborght et al. [29] found that using a social robot for storytelling significantly reduced anxiety among autistic children. While AI-based social robots, such as Nao, Probo, and Moxie, are available on the consumer market, they often cost between \$1,500

Table 1: Thematic Analysis Results and Sample Verbatim

Triggers of meltdown	Communicate difficulty	Attempted solutions
Sensory overloads(loud noises, busy crowds, flashing lights etc.) and lack of control (changes to the plan)	Not knowing the cause and how to deal with the meltdown, Parents feel helpless	Provide tablet time, Identifying the triggers to avoid them, Redirecting attention
"My son, who has autism, had a meltdown in the store. We know that meltdowns are a common occurrence for children with autism, but that doesn't make it any less challenging to navigate."	"My son refuses to get ready and just lies down on the floor, completely shut down. I have tried putting him to bed earlier, making it a game in the morning. Nothing is working. I'm overwhelmed at this point and not sure what way to turn."	"Identifying your own triggers and understanding what they are, you can begin to develop your own blueprint on how to either avoid or minimize them" "When he is using the tablet with his educative apps he doesn't do it."

to \$3,000, and it can be difficult to customize the contents to meet the unique needs of children and caregivers [25].

To this end, our work aims to design an accessible mobile application that allows caregivers to easily tailor contents, such as stories and soothing music, to indirectly communicate with their children. This mobile application is designed to be carried anywhere by children and can be used to prompt communication when children are expected to experience stress. Our contribution lies in developing an affordable and customizable solution that can help caregivers manage emotional meltdowns in autistic children and improve their overall well-being.

3 PRELIMINARY STUDY: IDENTIFY DESIGN REQUIREMENTS

To better understand the ongoing issues faced by families with autistic children, we explored the social media and online community. Social media platforms provide a wealth of information that can be utilized to gain insights into the needs and concerns of the target audience and identify design requirements. Following the approach of Thoren et al [28] in using social media platforms for data collection, we analyzed the most recent wall posts in the largest public Facebook parent group for autistic children and Reddits. After reviewing hundreds of postings, four researchers selected and systematically analyzed 60 posts for final data analysis. In addition, we searched for videos on YouTube that effectively illustrate the challenges that caregivers face with autistic children. We used various keywords such as “autism,” “support,” “struggles,” and “transition” to search for videos on YouTube. Four researchers selected 12 videos that best represent the challenges faced by autistic children and summarized the contents of the video about the user, environment, goals, and pain points. We conducted a thematic analysis to examine the videos and textual data. We focused on identifying information that is pertinent to the current struggles and coping strategies of autistic children and their caregivers. The researchers independently underwent an open-coding process. Table 1 summarized the finding from thematic analysis. The results indicate that children’s meltdown

is one of the common challenges faced by parents and caregivers. The following insights from the results guided the design of Kodi:

- Design Requirement 1: Alleviate anxiety and promote coping mechanisms.
- Design Requirement 2: Provide multiple ways for children to express their emotions.
- Design Requirement 3: Allow customization and help caregivers connect with children
- Design Requirement 4: Identify triggers that can lead to meltdowns.

4 KODI:CHATBOT TO CONNECT CHILDREN AND THEIR CAREGIVERS

4.1 Design Process

Brainstorming, Scenario, Persona: To address the identified design requirements, we conducted a series of brainstorming sessions, followed by developing scenarios to better understand the target audience's perspective. Through this process, we created personas, including Sarah, an 11-year-old child with autism, and her father, John. Sarah is anxious about her upcoming school trip and struggles to communicate her emotions with her teachers. John is worried that Sarah may have a meltdown during the trip. He hopes to be able to comfort Sarah during the journey and help her homeroom teachers identify potential triggers to avoid similar situations.

Wireframe, Prototype: We began with low-fidelity sketches on whiteboards and paper, followed by a medium-fidelity mock-up (as shown in Figure 2), and developed interactive hi-fidelity prototypes using Figma. Kodi offers both a child’s view and a parent’s dashboard. For the development of Kodi, we plan to use the Anima plug-in, which facilitates the conversion of Figma files into react.js and vue.js.

Initial User Testing: To evaluate Kodi's usability, we conducted think-aloud sessions with 6 participants using UserTesting.com. These participants were between the ages of 18 and 50 and lived in the United States. They were asked to use multiple input modalities

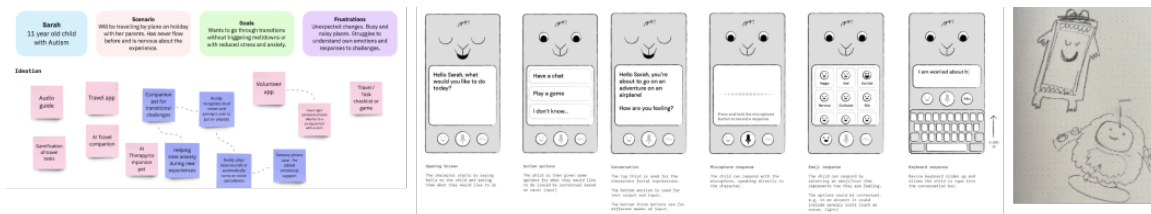


Figure 2: brainstorming on miro board (left), mid-fidelity wireframe (middle), sketches for the appearance (right)

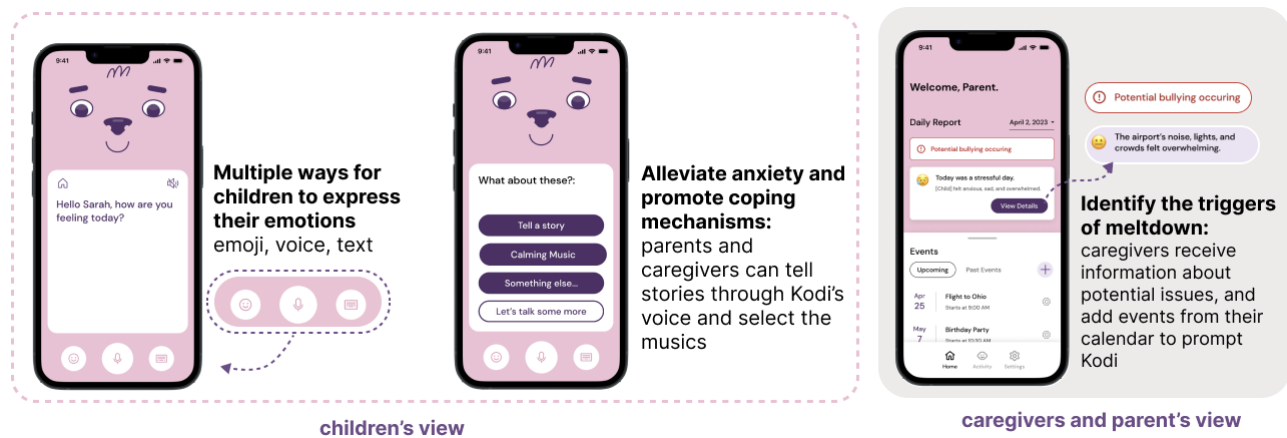


Figure 3: Kodi is a customizable chatbot designed to assist children with Autism Spectrum Disorder (ASD) in managing their emotions and expressing their feelings. Kodi offers both a child-oriented interface and a parent/caregiver dashboard. Through Kodi, parents and caregivers can communicate indirectly with their children by sharing stories and selecting music that can help to soothe them.

(emoji and text) to indicate their current emotions and complete the breathing exercise when feeling stressed. The feedback we received showed that Kodi was user-friendly and easy to use, making it an appropriate tool for children to express their emotions. Participants particularly enjoyed the change in Kodi's facial expressions and identified it as their favorite feature of the app.

4.2 Bringing Requirements to Life: Kodi

Alleviate anxiety and promote coping mechanisms: The thematic analysis results highlighted the significance of utilizing diverse coping strategies to manage children's meltdowns. Parents have found providing mobile tablets to be helpful in redirecting the child's attention during a meltdown, along with playing games and the child's favorite music and videos. Additionally, multiple studies have demonstrated the effectiveness of storytelling in redirecting the attention of autistic children and calming them down [10, 29]. Building upon these findings, Kodi features a range of coping mechanisms, including storytelling, breathing exercises, playing children's favorite music or videos, and mobile games. Kodi prompts children to provide information about their current emotional state and environment, and then suggest the most appropriate coping strategies based on contextual information and input.

Multiple ways for children to express their emotions: Providing various methods for children to express their emotions is crucial, as some children may find it difficult to articulate their feelings verbally. Many parents have reported challenges when their children refuse to communicate or shut down. Previous research has suggested that visual interaction, such as using picture cards and images, can be helpful in assisting autistic children to express their emotions [5, 7, 17]. In light of this, Kodi allows children to use various input methods, such as text, images, and speech, to communicate their current emotions and difficulties. This allows users to converse with Kodi in the way they feel most comfortable, and simultaneously teach and familiarize younger children with their own emotions.

Customization of appearance with haptic stimulation: In social media and online forums, parents have discussed the strong attachment that their children with ASD develop to specific objects such as trains, dinosaurs, and stuffed animals. Previous research has also suggested that individuals with ASD tend to form strong attachments to inanimate objects [11]. Furthermore, studies reveal that haptic feedback, such as the use of a furry blanket or other tactile stimulation, can be an effective way to calm individuals during a meltdown [9, 21]. For instance, Gringras et al. [8] found that haptic feedback from weighted blankets was effective in promoting relaxation and reducing anxiety among children with ASD. Building on these findings, we have integrated features into the Kodi app that allow for customization of its appearance, including the character's appearance, colors, and facial expressions. Specifically, Kodi's mobile phone cases can be designed to resemble a child's favorite character or animal, which can promote greater engagement and bonding between children and Kodi. By providing children and caregivers with the ability to personalize Kodi's appearance, we aim to facilitate easier communication, and to create a more comfortable environment for children to share their feelings and emotions.

Help parents and caregivers connect with children: One of the key features of Kodi is to facilitate communication between parents or caregivers and their children via a virtual chat-bot. With Kodi, parents or caregivers can create and customize stories with multiple voices, setting the context in which the story will be presented to their children. For example, one parent from a Facebook parents group shared how showing a video of a toy train has a soothing effect and redirects their child's attention during stressful moments. With Kodi, parents can easily upload pre-recorded audio files or personally narrate the stories using different voices. We will make use of open-source voice changing software like VoiceMed, which enables users to modify voices into chipmunk, robot, and cartoon characters. This will add an exciting and engaging element to the storytelling experience for both parents and children. Additionally,

parents or caregivers can customize content such as music and select children's favorite games, to better suit the individual needs of their children. This approach allows Kodi to deliver these specially tailored and personalized contents to their children, providing them with a unique and comforting experience.

Identify the triggers of meltdown: The findings from the thematic analysis indicate that parents recognize the importance of identifying the common triggers that cause their children to have a meltdown, such as being in crowded areas or bright light. Kodi will encourage children to express their current emotions and inquire about the factors causing stress, for example, asking questions like "What made you upset?" Kodi will carefully analyze contextual information, considering when, how, and where a child might experience a meltdown. It will then assist caregivers in recognizing the specific factors that trigger such behavior. Moreover, Kodi will also send alerts to parents and caregivers if potential threats arise, such as incidents of bullying at school.

5 NEXT STEPS AND FUTURE RESEARCH DIRECTION

In this study, we present Kodi, a mobile virtual conversational agent specifically designed to facilitate communication between autistic children and their caregivers during challenging situations. Previous research has demonstrated the effectiveness of social assistive robots and virtual agents in engaging autistic children in conversations and alleviating anxiety related to human interactions. However, commercially available assistive robots like Moxie tend to be costly and lack the flexibility to tailor their features to suit the unique needs of individual children [2, 3]. The primary contribution of this study is to establish the design requirements for developing a conversational chat-bot that caters to the needs of autistic children and their caregivers. While prior studies have primarily discussed using conversational chat-bots for speech therapy and social training for autistic children, our approach centers on utilizing a customizable conversational chat-bot as an intermediary between autistic children and their caregivers. This will enable caregivers to share stories with children through Kodi's voice and customize the coping tasks provided to the children (e.g., games, stories, musics), creating a more personalized and beneficial experience.

Our next step is to conduct a participatory design study with autistic children and their caregivers to refine the Kodi app's design. One crucial aspect to consider in Kodi's development is the privacy concerns of children. Currently, the app permits parents or caregivers to monitor their child's emotional struggles by receiving feedback from Kodi and information about potential issues. However, as the app's target audience is school-aged autistic children, this approach may raise concerns regarding privacy. Although Kodi can aid children by identifying triggers of anxiety and stress, future iterations of the app may require additional attention to ensure the child's awareness and consent regarding the feedback section. After gathering feedback from the participatory design sessions, our plan is to utilize an intelligent chatbot Application programming interface (API) to develop Kodi. Specifically, Kodi will be implemented using Botkit, react.js, and node.js. Future research can examine the long-term effects of using virtual social agents to manage emotional meltdowns in children. Additionally, the data

collected through Kodi will allow for the application of machine learning technology to predict and potentially prevent meltdowns, providing further support for parents and caregivers.

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